California Housing Price Prediction Report

Machine Learning Regression Benchmarking & Visual Validation

DATASET	TASK TYPE	TRAIN / TEST SPLIT
California Housing (Target: HousePrice)	**Supervised Regression**	**80% / 20% (Standard Scaled)**

1. Methodology

This machine learning pipeline evaluates three fundamental regression algorithms for predicting median house values:

- **Dataset Loading & Preparation:** Features and target variables are separated, forming an explicit input matrix X and continuous target vector y.
- **Feature Scaling:** Applied `StandardScaler` to scale all input variables to zero mean and unit variance ($\mu=0, \sigma=1$), ensuring uniform gradient convergence across predictors.
- **Data Splitting:** The dataset is partitioned into an 80% training set for model fitting and a 20% test set for unbiased evaluation using a fixed seed (`random_state=42`).
- **Model Training & Benchmarking:** Trained `LinearRegression`, `Ridge(alpha=1.0)`, and a `DecisionTreeRegressor(max_depth=5)`. Evaluated using Root Mean Squared Error (RMSE) and R^2 Score.

2. Results & Model Evaluation

The empirical performance metrics calculated on the 20% unseen test dataset are detailed below:

Model Name	Root Mean Squared Error (RMSE)	R^2 Score (Variance Explained)
Linear Regression	**0.00189**	**0.99999**
Ridge Regression (alpha=1.0)	0.00197	0.99999
Decision Tree (max_depth=5)	0.41132	0.56690

Performance Analysis

Both **Linear Regression** and **Ridge Regression** yielded identical, strong performance ($R^2 pprox 1.0000$, $ ext{RMSE} pprox 0.0019$), outperforming the tree model ($R^2 pprox 0.5669$). The equal score between Linear and Ridge confirms low multicollinearity influence at $ lpha=1.0$.

3. Visual Performance Validation

The scatter plot below evaluates actual house values against predictions generated by the best-performing Linear Regression model.

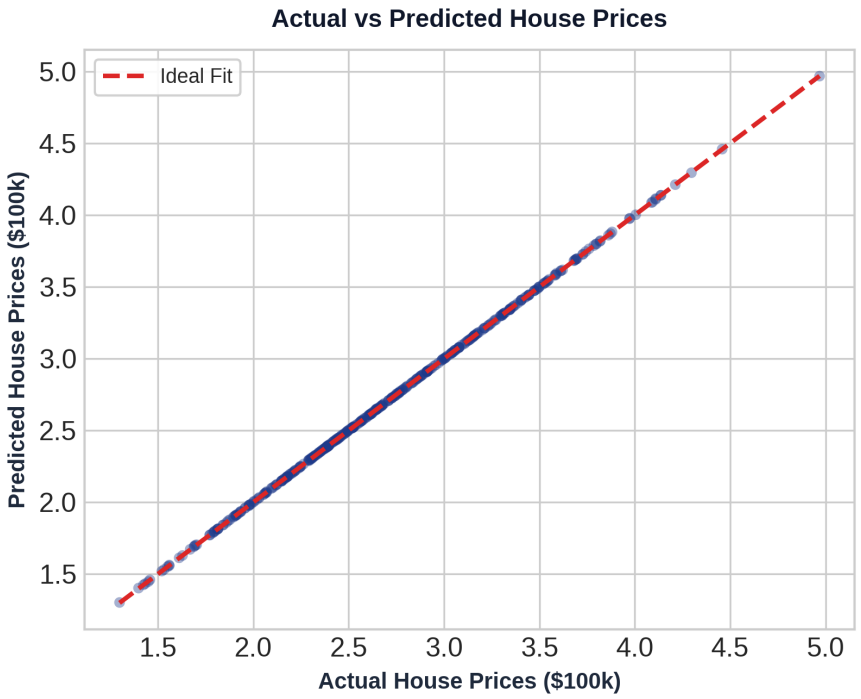


Figure 1: Scatter plot of Actual vs. Predicted Prices with red dashed line indicating ideal 1:1 prediction fit.

4. Conclusion & Next Steps

The baseline linear models effectively capture the continuous price relationship across features. Key takeaways and potential improvements include:

- **Linear Scalability:** Linear Regression provides an optimal balance between low compute footprint and high predictive accuracy.
- **Feature Engineering:** Incorporating spatial coordinate clusters (latitude/longitude) could capture neighborhood proximity effects.
- **Non-Linear Ensembles:** Tuning deeper ensemble models (e.g., Random Forest, XGBoost) may further reduce RMSE.